



process will centre-stage the community and will be provided momentum and sustenance through the collective efforts of all Government agencies and Non-Governmental Organisations (NGOs).

The Disaster Management Act 2005 has not only created the institutional mechanisms like Disaster Management Authorities at the national, state and district levels, but also provided the foundation for a sound financial management strategy through the provisions for the establishment of Disaster Response Funds at the national, state and district levels as well as Disaster Mitigation Funds at the national, state and district levels, which are being considered by the Thirteenth Finance Commission. The Disaster Management Act 2005 also stipulated the establishment of a dedicated agency called the National Disaster Response Force (NDRF) consisting of personnel from the four para military forces (BSF, CISF, CRPF and ITBP) for assisting the district administrations and state governments in various aspects of disaster management like community preparedness, capacity building, mitigation efforts and emergency response. Eight battalions of the NDRF have been deployed in various strategic locations of the country and they have been trained and equipped to respond to natural disasters and man-made disasters, including Chemical, Biological, Radiological and Nuclear (CBRN) Emergencies.

The Disaster Management Act 2005 also laid the foundation for the setting up of the National Institute of Disaster Management (NIDM) as the nodal training institute in the country, coordinating the capacity building of various stakeholders in the field of disaster management through several state level training institutes.

NDMA's Earthquake Management Guidelines

The mandate of the National Disaster Management Authority (NDMA), as specified in the Disaster Management Act 2005, inter alia, has stipulated that NDMA can draw up the Guidelines for the effective management of natural and man-made disasters in the country. Given the high seismic risk and vulnerability in India, NDMA considered it a high priority to identify the critical gaps in the management of earthquakes in India and to work towards the design of appropriate strategies to address these critical gaps by bringing together various stakeholders, earthquake management experts, senior administrators, academics and disaster management practitioners. A Core Group of Experts was set up after the deliberations of a National Conference on Management of Earthquakes in India to prioritise the various steps that need to be taken to strengthen earthquake preparedness, introduce earthquake mitigation initiatives and improve the earthquake response capacity in the country. A series of workshops were held and meetings of the Core Group held to prepare the Draft Guidelines which were reviewed in regional conferences in Indian

Institute of Technology, Kanpur and Indian Institute of Technology, Bombay. The Guidelines on the Management of Earthquakes were finalized on the basis of the feedback received from the government officials and earthquake management specialists.

On 16th May 2007, Shri Shivraj Patil, Hon'ble Union Home Minister of India, in the presence of Shri Kapil Sibal, Hon'ble Minister for Earth Sciences and Science and Technology, released the National Disaster Management Guidelines for the Management of Earthquakes. These Guidelines, prepared with the active support of more than 300 seismic experts in the country, identified the following critical concerns in the management of earthquakes in India.

- Lack of awareness among various stakeholders about earthquake risk and vulnerability
- Inadequate attention to structural mitigation measures in the engineering education syllabus
- Inadequate monitoring and enforcement of earthquake-resistant building codes and town planning bye-laws
- Absence of systems of licensing of engineers and masons
- Absence of earthquake-resistant features in non-engineered construction in suburban and rural areas
- Lack of formal training among professionals in earthquake-resistant construction practices; and
- Lack of adequate preparedness and response capacity among various stakeholder groups.

The weak compliance and enforcement of Building Codes has often been attributed to the prevailing impression that the earthquake-resistant building codes prepared by the Bureau of Indian Standards are mostly recommendatory. The Earthquake Management Guidelines has now made it mandatory for the strict compliance and enforcement of earthquake-resistant building codes and other safety related codes, as well as the modified techno-legal regime, consisting of revised town planning bye-laws, land-use zoning and Development Control Regulations, for all new construction with immediate effect in towns and cities in zones III, IV and V.

Six Pillars of Earthquake Management in India

The Earthquake Management Guidelines have identified the need to evolve a strategy based on the following six pillars.

- Earthquake-resistant construction of all new structures
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- Selective seismic strengthening and retrofitting of existing priority structures and lifeline structures

